

Synaptic Depression Modeling

Data generation and model fitting

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Abstract

We describe a real-time adaptive Bernoulli state-space model (BSSM) for a network of interacting neurons with short-term synaptic depression (STD) and a lag- M synaptic memory kernel. The STD mechanism is represented as a latent dynamical resource variable that is updated at each discrete time step, coupled to observed binary spike indicators through a Bernoulli likelihood. For context, we also briefly review the Tsodyks–Markram (TM) resource model, which provides the continuous-time biological foundation for the discrete STD dynamics used in the BSSM.

1 Short-Term Synaptic Depression (STD) in Cortical Synapses

Short-term synaptic depression (STD) is a fast, reversible decrease in synaptic strength that occurs when a presynaptic cortical neuron fires repeatedly over short time windows (typically from milliseconds to seconds).

Activity-decaying phenomenon. When presynaptic firing is high, the postsynaptic response (EPSCs or IPSCs) becomes progressively smaller for successive spikes, producing a decaying envelope of synaptic efficacy during bursts. After presynaptic activity pauses, synaptic strength gradually recovers to baseline levels.

Main biological mechanism. STD is commonly modeled as depletion of synaptic “resources” required for effective neurotransmitter release. At each presynaptic spike, release depends on (i) the availability of readily releasable vesicles (or release-ready sites) and (ii) an effective probability of release (often related to intracellular calcium dynamics). Sustained activity depletes resources and reduces future release, while periods of silence allow recovery.

Relevant timescales. Recovery is typically on intermediate timescales (~ 100 ms to a few seconds), depending on synapse type and experimental protocol.

Functional consequences. STD acts as a rate-dependent synaptic filter: it attenuates sustained high-rate input, emphasizes transient changes in firing, and contributes to network stabilization by preventing runaway excitation.

2 Computational Models with STD

We describe two STD frameworks. The TM model (Section 2.1) is included for conceptual context only. The BSSM (Section 2.2) is the target implementation model.

The following notation is shared across both models: $U \in (0, 1]$ is the utilization parameter (fraction of available resources used per spike), and τ_{rec} is the STD recovery time constant.

Tsodyks–Markram short-term synaptic depression model

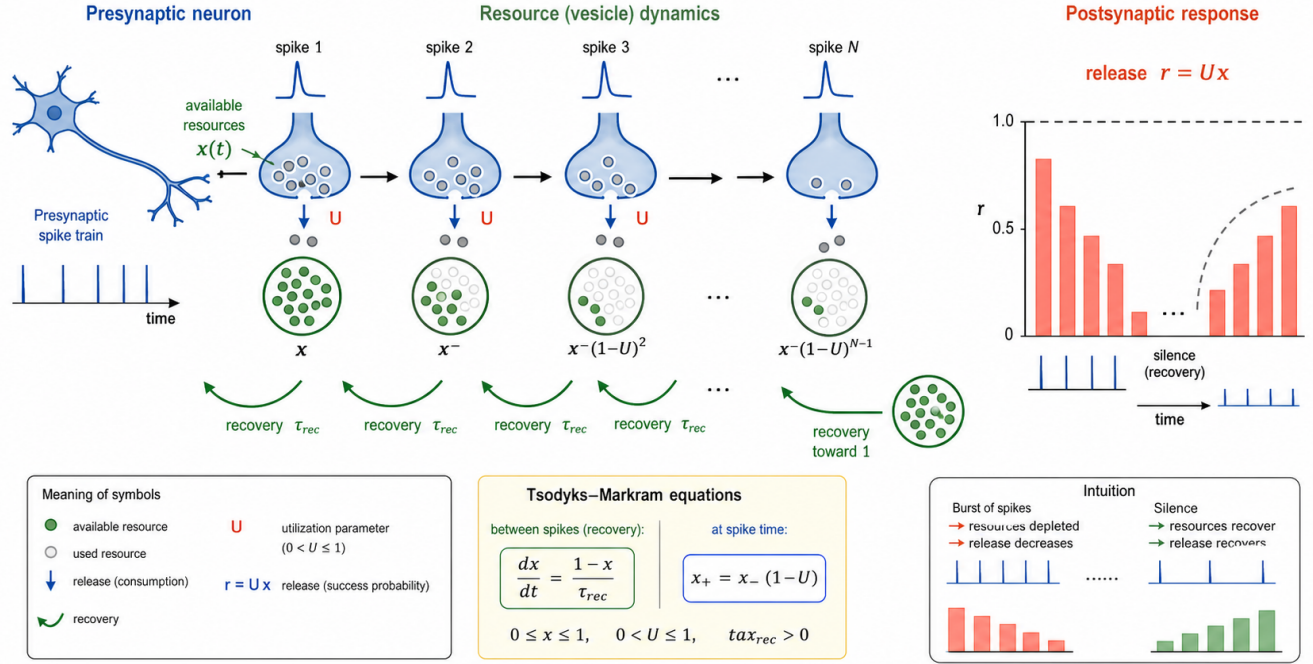


Figure 1: Cartoon of the Tsodyks–Markram STD resource model. Presynaptic spikes consume a fraction U of the available resource x , making successive release events smaller; during silence, the resource recovers toward $x = 1$ with time constant τ_{rec} .

2.1 Tsodyks–Markram (TM) Resource Model: Conceptual Background

The *Tsodyks–Markram (TM)* model [1] is a widely used continuous-time resource model for *short-term plasticity (STP)*. In the STD-only version, the synapse has a single *available-resource* variable $x(t) \in [0, 1]$. Between presynaptic spikes, resources recover exponentially:

The cartoon in Fig. 1 summarizes the STD-only TM mechanism: spikes multiply the available resource by $(1 - U)$, while inter-spike intervals allow the resource to recover toward one.

Between presynaptic spikes,

$$\frac{dx}{dt} = \frac{1-x}{\tau_{\text{rec}}}. \quad (1)$$

At a presynaptic spike time $t = t_k$, a fraction U of available resources is consumed:

$$x(t_k^+) = x(t_k^-) (1 - U), \quad (2)$$

and the released neurotransmitter amplitude at that spike is

$$r(t_k) = U x(t_k^-). \quad (3)$$

The TM model provides the biological motivation for the discrete STD resource update used in the BSSM below (Eq. 5), which is the exact discretization of Eq. (1) combined with Eq. (2).

Interpretation of U . The utilization parameter U is a fraction of the resources available at the moment of a spike, not a fixed fraction of the original full pool. For example, if $U = 0.5$ and several spikes arrive with negligible recovery between them, a fully recovered synapse releases 0.5 of the full pool on the first spike, then 0.25 on the second, then 0.125 on the third, and so on. Thus the axon can still generate more than two closely spaced spikes; STD only makes the effective synaptic release decrease geometrically during the burst.

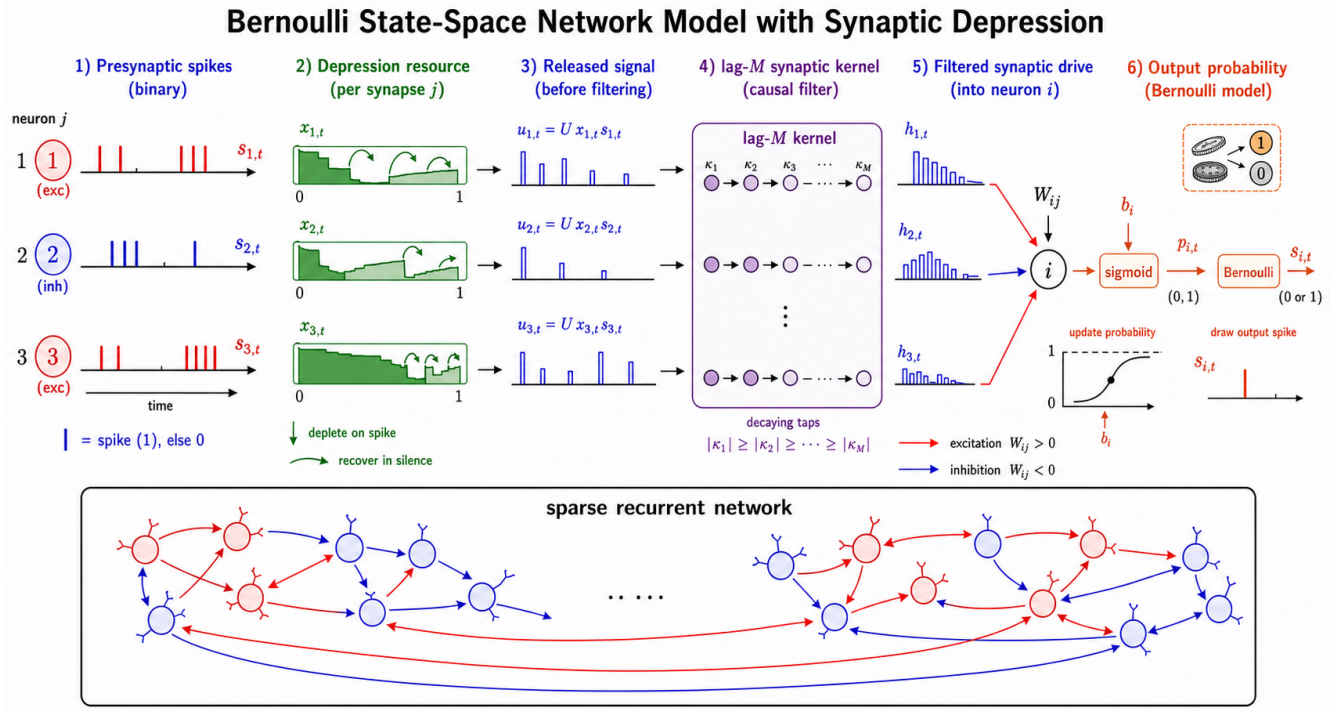


Figure 2: Cartoon of the full BSSM with STD and a lag- M synaptic kernel. Presynaptic binary spikes $s_{j,t}$ modulate the latent resource $x_{j,t}$, produce STD-weighted release $u_{j,t}$, pass through a causal synaptic kernel to form filtered drive $h_{j,t}$, and enter the postsynaptic Bernoulli observation model through signed recurrent weights W_{ij} and baseline b_i .

2.2 Real-Time Adaptive Bernoulli State-Space Network Model (BSSM) with STD and Lag- M Kernel

The *Bernoulli state-space model (BSSM)* represents a network of N neurons using (i) latent dynamical states encoding STD resources and (ii) a Bernoulli observation model for binary spike indicators conditioned on those states. This is the primary model of interest. At physiological firing rates (30–50 Hz) and bin width ($\Delta t = 1$ ms), the per-bin spike probability is $p \approx r\Delta t = 0.03$ – 0.05 . The corresponding Poisson probability of two or more spikes in one bin is approximately $p^2/2$, ranging from 4.5×10^{-4} to 1.25×10^{-3} . Thus the Bernoulli model is a good approximation for binary spike indicators while explicitly enforcing at most one spike per bin.

Figure 2 summarizes the sequential data-flow view of the model: the recurrent weighted input determines the postsynaptic spike probability, sampled presynaptic spikes generate release from the current STD resource, and that release updates both the next STD state and the filtered drive available in the next bin.

Discrete time and notation. Let $t \in \{0, 1, \dots, T-1\}$ index time bins of width Δt . For each neuron j , the primary observable is the binary *spike indicator*

$$s_{j,t} = \begin{cases} 1 & \text{neuron } j \text{ fired a spike in bin } t, \\ 0 & \text{neuron } j \text{ was silent in bin } t, \end{cases} \quad (4)$$

which records only *whether* a spike occurred. The downstream efficacy of that spike is determined separately by the latent STD resource variable $x_{j,t} \in [0, 1]$, which encodes the fraction of neurotransmitter available for release at the start of bin t (see Eq. 5). A spike that occurs when resources are depleted

($x_{j,t} \approx 0$) contributes little to postsynaptic drive, whereas a spike after full recovery ($x_{j,t} \approx 1$) contributes maximally.

STD latent state dynamics. Each presynaptic neuron j has its own latent resource variable $x_{j,t}$, tracking the STD state of that neuron’s outgoing synapses independently of all other neurons. The full network state is the collection $\{x_{j,t}\}_{j=1}^N$, with each variable evolving as the exact discretization of the TM continuous-time recovery (Eq. 1) combined with spike-triggered consumption (Eq. 2):

$$\begin{aligned} x_{j,t}^{\text{dep}} &= \underbrace{x_{j,t}(1 - U s_{j,t})}_{\text{spike-triggered depletion}}, \\ x_{j,t+1} &= \underbrace{1 - (1 - x_{j,t}^{\text{dep}})e^{-\Delta t/\tau_{\text{rec}}}}_{\text{exponential recovery after depletion}}, \quad j = 1, \dots, N. \end{aligned} \quad (5)$$

Release during bin t uses the pre-spike resource $x_{j,t}$. If a spike occurs, a fraction U is first consumed, and the depleted resource then recovers toward 1 over the interval to bin $t + 1$. Because the update depends only on neuron j ’s own spike indicator, resource variables of different neurons are conditionally independent given their respective spike histories, allowing all N updates to proceed in parallel.

STD-modulated release. The effective synaptic release from neuron j in bin t is:

$$u_{j,t} = U x_{j,t} s_{j,t}. \quad (6)$$

This quantity is zero in the absence of a spike, small when resources are depleted by recent activity, and large (up to U) after full recovery.

Lag- M synaptic kernel. Synaptic currents have temporal extent beyond a single bin. We convolve the STD-modulated release with a causal lag- M kernel $\kappa = (\kappa_1, \dots, \kappa_M)^\top$, $\kappa_\ell \geq 0$, encoding how a release event at lag ℓ bins contributes to postsynaptic drive at time t . The effective STD-filtered synaptic input from neuron j at time t is:

$$h_{j,t} = \sum_{\ell=1}^M \kappa_\ell u_{j,t-\ell}. \quad (7)$$

Here $M\Delta t$ is the finite lag-history span for the fast synaptic current. For an exponential synaptic kernel, practical choices take $M\Delta t$ to be several synaptic time constants τ_s so that the current tail is small at the truncation boundary. This lag window is distinct from the STD recovery memory: the slower recovery governed by τ_{rec} is carried by the latent state $x_{j,t}$ rather than by the finite synaptic kernel. Practical kernel choices include exponential decay, raised cosine basis functions, or a free learnable filter (see Section 3.2 for the generator choice).

Bernoulli observation model. For each postsynaptic neuron i , the conditional spiking probability is:

$$p_{i,t} = \sigma \left(b_i + \sum_{\substack{j=1 \\ j \neq i}}^N W_{ij} h_{j,t} \right), \quad (8)$$

where $\sigma(z) = 1/(1 + e^{-z})$ is the logistic sigmoid, b_i is a baseline log-odds for neuron i , and W_{ij} is the synaptic weight from neuron j to neuron i (positive for excitatory, negative for inhibitory connections). Self-connections are excluded by the constraint $j \neq i$. The spike indicator is then drawn as:

$$s_{i,t} \sim \text{Bernoulli}(p_{i,t}). \quad (9)$$

For $t - \ell < 0$, pre-history values $u_{j,t-\ell}$ are set to zero, corresponding to a fully recovered synapse with no prior spiking.

Summary of model parameters. The full parameter set of the BSSM is $\theta = \{b_i, W_{ij}, \kappa_\ell, U, \tau_{\text{rec}}\}$, where b_i and W_{ij} are neuron- and synapse-specific, while this model assumes a single shared utilization parameter U for all presynaptic neurons and outgoing synapses. The recovery time τ_{rec} is also treated as shared in the generator and estimation procedure below. In the generator (Section 3), all parameters are fixed to known ground-truth values; in inference, they constitute the quantities to be estimated.

Real-time and adaptive aspects. The term *real-time* refers to the fact that both the latent STD state $x_{j,t}$ and parameter estimates can be updated sequentially as new spike indicators $s_{i,t}$ arrive, without reprocessing the full data history. The term *adaptive* refers to the property that $x_{j,t}$ automatically tracks recent presynaptic spiking through Eq. (5): resources deplete during bursts and recover during silence, so the effective synaptic gain $u_{j,t}$ adapts online to the presynaptic firing history. Together, these properties make the BSSM suitable for closed-loop neural decoding or real-time inference scenarios.

3 Data Generator: Cortical Network with STD

To validate the BSSM and test inference procedures, we generate synthetic spike trains from a network that mimics key properties of mammalian cortical dynamics. The generator uses the same BSSM structure as the inference model (Section 2.2), but with all parameters fixed to known ground-truth values, enabling direct comparison between true and estimated quantities.

3.1 Network Architecture

We consider a network of $N \in [50, 500]$ neurons partitioned following Dale’s law into $N_E = 0.8N$ excitatory and $N_I = 0.2N$ inhibitory neurons. The code constructs the initial synaptic matrix $W \in \mathbb{R}^{N \times N}$ with a spatial connections Dale generator. Neurons are placed on a two-dimensional grid over $[0, L] \times [0, H]$ with minimum spacing $d_{\min}$, and their order is sorted by position. For each presynaptic neuron j , an out-degree k_j is drawn from the configured range and k_j distinct postsynaptic targets are sampled without replacement with probability proportional to $D_{ij}^{-\delta}$, excluding $i = j$. This gives nearby neurons higher connection probability while keeping the diagonal zero.

The nonzero weights obey Dale’s law column-by-column:

$$W_{ij}^{(0)} = \begin{cases} +a_{ij} & \text{if neuron } j \text{ is excitatory,} \\ -\frac{N_E}{N_I} a_{ij} & \text{if neuron } j \text{ is inhibitory,} \end{cases} \quad a_{ij} \sim \text{Uniform}(1 - v, 1 + v), \quad (10)$$

for sampled edges only, with $W_{ii}^{(0)} = 0$. Finally, the whole matrix is rescaled to the requested spectral radius R ,

$$W = W^{(0)} \frac{R}{\rho(W^{(0)})}, \quad (11)$$

which preserves sparsity, signs, Dale’s law, and the zero diagonal.

3.2 Synaptic Kernel

We use an *exponential decay kernel* as the ground-truth synaptic filter:

$$\kappa_\ell = (1 - \alpha) \alpha^{\ell-1}, \quad \ell = 1, \dots, M, \quad (12)$$

where $\alpha = e^{-\Delta t / \tau_s}$ and τ_s is the synaptic time constant. This kernel corresponds to a standard single-compartment synaptic current model and is the exact discrete-time analog of the continuous-time shot-noise synaptic model. It is normalized so that $\sum_{\ell=1}^M \kappa_\ell \approx 1$ for $M\Delta t \gg \tau_s$.

A key computational advantage of this choice is that the lag- M convolution in Eq. (7) can be evaluated by a tail-corrected recursive update:

$$h_{j,t+1} = \alpha h_{j,t} + (1 - \alpha) u_{j,t} - (1 - \alpha) \alpha^M u_{j,t-M}, \quad (13)$$

with $u_{j,t-M} = 0$ when $t - M < 0$. This recurrence is exactly equivalent to the finite lag- M convolution for the exponential kernel and avoids explicitly summing over all M lags at every time step. The decay rate of κ_ℓ is set by the fast synaptic time constant τ_s , whereas the slower STD time constant τ_{rec} enters through the latent state update in Eq. (5).

3.3 Ground-Truth Parameter Values

Table 1 lists the ground-truth parameter values used in the generator, motivated by mammalian cortical physiology.

Parameter	Symbol	Value	Rationale
Time bin width	Δt	1 ms	Resolves spikes; keeps p small
Synaptic time constant	τ_s	3–5 ms	AMPA-like fast excitation
STD recovery time	τ_{rec}	200–500 ms	Typical cortical STD
Utilization parameter	U	0.3–0.5	Shared moderate depression
Weight scale	v, R	$v = 0.4, R = 0.90$	Spectral-radius scaling
Baseline log-odds	b_i	sampled idle-rate range	User-adjusted baseline drive
Kernel length	M	$\gtrsim 3\text{--}5 \tau_s / \Delta t$	Fast synaptic lag window

Table 1: Ground-truth generator parameters. Ranges reflect variability across synapse types and network sizes. The kernel length M controls the finite lag history for the fast synaptic filter and is separate from the STD recovery time τ_{rec} . The baseline b_i is expressed as a log-odds consistent with the Bernoulli observation model (Eq. 8).

3.4 Generation Procedure

Given the fixed parameters, synthetic spike indicators are generated sequentially for $t = 0, 1, \dots, T - 1$ as follows.

Algorithm 1 Synthetic network spike train generator

- 1: **Initialize:** $x_{j,0} \leftarrow 1$ for all j (fully recovered); $h_{j,0} \leftarrow 0$ and $u_{j,t} \leftarrow 0$ for $t < 0$; draw $W, \{b_i\}$ as in Eqs. (10)–(11).
 - 2: **for** $t = 0, 1, \dots, T - 1$ **do**
 - 3: **for** each neuron $i = 1, \dots, N$ **do**
 - 4: Compute spiking probability: $p_{i,t} \leftarrow \sigma(b_i + \sum_{j \neq i} W_{ij} h_{j,t})$ (Eq. 8)
 - 5: Draw spike indicator: $s_{i,t} \sim \text{Bernoulli}(p_{i,t})$ (Eq. 9)
 - 6: **end for**
 - 7: **for** each neuron $j = 1, \dots, N$ **do**
 - 8: Compute STD-modulated release from pre-spike resources: $u_{j,t} \leftarrow U x_{j,t} s_{j,t}$ (Eq. 6)
 - 9: Update STD resource: $x_{j,t+1} \leftarrow 1 - (1 - x_{j,t}(1 - U s_{j,t}))e^{-\Delta t / \tau_{\text{rec}}}$ (Eq. 5)
 - 10: Update recursive synaptic filter: $h_{j,t+1} \leftarrow \alpha h_{j,t} + (1 - \alpha) u_{j,t} - (1 - \alpha) \alpha^M u_{j,t-M}$ (Eq. 13)
 - 11: **end for**
 - 12: **end for**
 - 13: **Output:** spike indicators $\{s_{j,t}\}$, ground-truth STD states $\{x_{j,t}\}$, weight matrix W .
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3.5 Network Stability Check

With recurrent excitatory connections, the network can exhibit runaway activity if synaptic weights are too large. Before generating long recordings, we recommend a short pilot run (~ 1 s) and verification that mean firing rates across neurons remain in the target range of 30–50 Hz. If rates exceed this range, one should reduce the spectral-radius target R or increase U : stronger STD provides a natural activity-dependent stabilizing mechanism by reducing synaptic efficacy during high-rate episodes.

4 Initializing U from Raw Spike Data

The utilization parameter U controls the multiplicative depletion of synaptic resources at each presynaptic spike. Because this depletion affects the relative release generated by consecutive spikes from the same neuron, tight-pair statistics provide a useful initializer for the network-wide U . In this section, U denotes the true or model utilization, while $\tilde{U}$ denotes an inferred value obtained from data. This raw estimate is not treated as final: it is refined later by likelihood optimization together with τ_{rec} and the recurrent parameters. We do not attempt to pre-estimate τ_{rec} from raw consecutive-spike ISIs: at the target 30–50 Hz rates, typical ISIs are tens of milliseconds, much shorter than the expected 0.2–0.5 s recovery time. Instead, τ_{rec} is treated as unknown and initialized from a fixed physiological guess in the full fitting procedure. This section estimates the recording requirements for the raw $\tilde{U}$ initializer.

4.1 The Burst-Pair Estimator for U

Consider two consecutive spikes from presynaptic neuron j , separated by an inter-spike interval (ISI) δ . Let x_1^- be the resource immediately before the first spike. After spike-triggered depletion and recovery toward one, the resource immediately before the second spike is

$$x_2^- = 1 - (1 - x_1^-(1 - U)) e^{-\delta/\tau_{\text{rec}}}. \quad (14)$$

For this pair, define the two release amplitudes by the same rule as Eq. (6): $u_1 = Ux_1^-$ for the first spike and $u_2 = Ux_2^-$ for the second spike. The ratio of second-spike to first-spike release is therefore

$$\rho(\delta) = \frac{u_2}{u_1} = (1 - U)e^{-\delta/\tau_{\text{rec}}} + \frac{1 - e^{-\delta/\tau_{\text{rec}}}}{x_1^-}. \quad (15)$$

In the linear-response regime, $\rho(\delta)$ can be estimated from the ratio of postsynaptic excess firing probabilities above baseline in the bins following the first and second presynaptic spikes. The key point is that the zero-ISI limit is independent of the unknown starting resource:

$$\lim_{\delta \rightarrow 0^+} \rho(\delta) = 1 - U. \quad (16)$$

Thus the estimator does not require a long silent period before the tight pair. It requires the pair itself to be sufficiently tight that recovery during the ISI is negligible. For finite δ , the recovery correction depends on the unknown x_1^- ; depleted states amplify the bias. A practical approximation is to replace x_1^- by the stationary mean resource expected at the observed presynaptic firing rate r :

$$\bar{x}(r, U, \tau_{\text{rec}}) \approx \frac{1}{1 + Ur\tau_{\text{rec}}}, \quad (17)$$

which follows by balancing recovery toward one against average spike-triggered depletion. Substituting $\bar{x}$ into Eq. (15) gives an average finite-ISI correction. To make the bias explicit, define the naive zero-ISI inferred value $\tilde{U}_0(\delta) = 1 - \hat{\rho}(\delta)$. The finite-ISI effect is an additive bias in the inferred utilization, not a multiplicative rescaling. Its magnitude is approximately

$$B_U(\delta) \equiv \left| \tilde{U}_0(\delta) - U \right| \approx \left(1 - e^{-\delta/\tau_{\text{rec}}} \right) \left[\frac{1}{\bar{x}(r, U, \tau_{\text{rec}})} - (1 - U) \right] \sim \frac{1 - e^{-\delta/\tau_{\text{rec}}}}{\bar{x}(r, U, \tau_{\text{rec}})}, \quad (18)$$

so high firing rates, which reduce $\bar{x}$, require tighter ISI windows. Let ε_U denote the allowed additive bias in the inferred utilization $\tilde{U}$. In continuous time, keeping this finite-ISI correction below ε_U would require an ideal small-window cutoff

$$\delta_{\text{ideal}} \approx \frac{\varepsilon_U \tau_{\text{rec}}}{\frac{1}{\bar{x}(r, \tilde{U}, \tau_{\text{rec}})} - (1 - U)}, \quad (19)$$

valid when $\delta_{\text{ideal}} \ll \tau_{\text{rec}}$. With 1 ms binned spike trains, however, the smallest resolvable nonzero ISI is one time bin. Therefore the practical tight-pair cutoff used for initialization is

$$\delta_{\text{max}} = \Delta t = 1 \text{ ms}. \quad (20)$$

This cutoff is not chosen because it makes the finite-ISI bias vanish; it is the finest window available in the sampled data. The typical ISI, $1/r \approx 20\text{--}33$ ms, is much too large to use as a tight-pair cutoff because recovery during that interval is no longer negligible. Because $\bar{x}$ itself depends on the unknown U , this raw estimator is naturally iterative in $\tilde{U}$: start from a nominal value (for example $\tilde{U} = 0.5$), compute $\bar{x}(r, \tilde{U}, \tau_{\text{rec}})$ and δ_{max} , estimate a new $\tilde{U}$ from the selected tight pairs, then update $\bar{x}$ and repeat until the initializer stabilizes.

With a nominal or current estimate of τ_{rec} and an average resource estimate $\bar{x}$, each observed ratio can be corrected before averaging:

$$\tilde{U}_{\text{corrected}}(\delta, \bar{x}) = 1 - e^{\delta/\tau_{\text{rec}}} \left[\hat{\rho}(\delta) - \frac{1 - e^{-\delta/\tau_{\text{rec}}}}{\bar{x}} \right]. \quad (21)$$

The tight-pair window is still useful after this correction because it limits sensitivity to model mismatch, refractory effects, and finite-sample noise in the excess-probability ratio. In the protocol below, the raw tight-pair estimate is used only to initialize $\tilde{U}$; the final value is obtained by likelihood optimization.

Treatment of τ_{rec} during initialization. The tight-pair initializer requires a value of τ_{rec} in Eqs. (17)–(21), but raw spike pairs do not give a practical standalone estimate of this slow time constant at the target rates. Therefore the initialization uses a fixed physiological guess, for example $\tilde{\tau}_{\text{rec}}^{(0)} = 0.4$ s. This value is not assumed to be correct. It only sets the initial resource scale and tight-pair window; the three-block likelihood optimization later updates $\tilde{\tau}_{\text{rec}}$ directly in Block C.

4.2 Statistical Requirements: Tight Pairs and Shots

Tight pairs are consecutive spike pairs from the same neuron with ISI $\delta \leq \delta_{\text{max}}$. For a Poisson spike train with rate r observed for duration $T_{\text{obs}} = T\Delta t$, the expected number of tight pairs per neuron is

$$\mathcal{P}_{\text{pairs}} = rT_{\text{obs}} \left(1 - e^{-r\delta_{\text{max}}} \right) \approx r^2 T_{\text{obs}} \delta_{\text{max}}, \quad (22)$$

with the approximation valid only when $r\delta_{\text{max}} \ll 1$. Thus the physical recording duration, firing rate, and coincidence window determine the pair count; the time bin width matters only insofar as it must resolve the selected ISIs.

Shots are the postsynaptic spike observations that follow selected tight-pair events. Since the model uses one shared U , tight pairs pool across all presynaptic neurons. If the mean number of true postsynaptic targets per presynaptic neuron is k , the approximate number of informative shots is

$$N_{\text{shots}}^{\text{eff}} \approx N \mathcal{P}_{\text{pairs}} k. \quad (23)$$

The raw data contain observations from all possible postsynaptic neurons, but only the connected targets contribute coherent signal to the excess-probability ratio. A rough precision target is $N_{\text{shots}}^{\text{eff}} \gtrsim 1/\varepsilon_U^2$. For $N = 500$ and $k = 75$, even $\mathcal{P}_{\text{pairs}} = 10$ per neuron gives 3.75×10^5 effective shots, so the estimator is usually limited more by the recovery-bias window than by shot noise.

4.3 Parameter Dependence and Operating Points

With the stationary-resource approximation, the ideal continuous-time coincidence window is set by the additive-bias tolerance in Eq. (19). In the actual 1 ms binned data, sub-millisecond windows cannot be resolved, so the practical choice is fixed to $\delta_{\max} = 1$ ms. Higher firing rates reduce $\bar{x}$ and therefore increase the implied finite-ISI bias, but they also increase the number of available tight pairs. The typical ISI, $1/r$, should not be used as a tight-pair cutoff: for $r = 30$ Hz it is about 33 ms, which would introduce an additive bias of order 0.5 in $\tilde{U}$ under the baseline parameters.

Table 2: Practical tight-pair statistics for true/nominal $U = 0.5$, $\delta_{\max} = \Delta t$, $N = 500$, mean out-degree $k = 0.15N = 75$, and fixed recording duration $T_{\text{obs}} = 1000$ s. $B_U(\delta)$ is the implied additive finite-ISI bias from Eq. 18, evaluated at $\delta = \Delta t$; for the 1 ms rows this is $B_U(1 \text{ ms})$. $\mathcal{P}_{\text{pairs}}$ uses the exact Poisson expression in Eq. 22. The resource mean $\bar{x}$ is computed from Eq. 17. Baseline row in bold.

Row	Description	Δt (ms)	r (Hz)	τ_{rec} (s)	$\bar{x}$	$B_U(\Delta t)$	$\mathcal{P}_{\text{pairs}}/\text{neuron}$
1	Baseline	1	30	0.4	0.143	0.016	887
2	Baseline, finer bin	0.5	30	0.4	0.143	0.008	447
3	Higher target rate	1	50	0.4	0.091	0.026	2,439
4	Higher target rate, finer bin	0.5	50	0.4	0.091	0.013	1,235
5	Short τ_{rec}	1	30	0.2	0.250	0.017	887
6	Short τ_{rec} , finer bin	0.5	30	0.2	0.250	0.009	447
7	Long τ_{rec}	1	30	0.5	0.118	0.016	887
8	Long τ_{rec} , finer bin	0.5	30	0.5	0.118	0.008	447

The main conclusion from Table 2 is that, for the $N = 500$ network considered here, network-wide pooling gives ample effective shots throughout the 30–50 Hz target firing range even when only one-bin tight pairs are used. The cost is a percent-level finite-ISI bias in the raw $\tilde{U}$ initializer. This is acceptable for initialization because the three-block likelihood fit subsequently refines $\tilde{U}$; it would not be acceptable as a final standalone estimator.

5 Summary of the Parameter Estimation Method

The true BSSM parameter set is $\theta = \{b_i, W_{ij}, \tau_{\text{rec}}, U, \tau_s\}$. The corresponding estimated parameter set is denoted $\tilde{\theta} = \{\tilde{b}_i, \tilde{W}_{ij}, \tilde{\tau}_{\text{rec}}, \tilde{U}\}$, with τ_s assumed known from the fast synaptic time scale or fixed by the chosen kernel. The utilization estimate $\tilde{U}$ is initialized from tight pairs using a nominal recovery time, while $\tilde{\tau}_{\text{rec}}$ is treated as unknown and initialized from a fixed physiological guess, $\tilde{\tau}_{\text{rec}}^{(0)} = 0.4$ s; neither is fixed thereafter. The final estimate is obtained by a three-block coordinate-descent maximum-likelihood procedure over $\{\tilde{b}_i, \tilde{W}_{ij}\}$, $\tilde{U}$, and $\tilde{\tau}_{\text{rec}}$.

5.1 Pre-processing: Initializing $\tilde{\tau}_{\text{rec}}$ and Estimating $\tilde{U}$

Step 1 — Initialize $\tilde{\tau}_{\text{rec}}$. Do not pre-estimate τ_{rec} from raw consecutive-spike ISIs. At 30–50 Hz, typical ISIs are only 20–33 ms, so they do not span the 0.2–0.5 s recovery range needed for a stable standalone estimate. Set a physiological starting value,

$$\tilde{\tau}_{\text{rec}}^{(0)} = 0.4 \text{ s.} \quad (24)$$

This value is only an initialization for the forward STD recursion and for the tight-pair correction; it is updated later in Block C.

Step 2 — Initialize $\tilde{U}$ iteratively. Estimate $\tilde{U}^{(0)}$ from the tight-pair release ratio in Eq. (15), pooling pairs across all presynaptic neurons. Because the finite-ISI correction depends on the typical pre-pair resource level, and that resource level depends on U , this initialization is a fixed-point procedure rather than a one-pass calculation. Start from a nominal value, for example $\tilde{U} = 0.5$; compute $\bar{x}(r, \tilde{U}, \tilde{\tau}_{\text{rec}}^{(0)})$ from Eq. (17); set the tight-pair window using Eq. (20); estimate a new $\tilde{U}$ using the corrected ratio in Eq. (21); then recompute $\bar{x}$ and repeat until the update changes negligibly. With an extremely narrow coincidence window, the zero-ISI approximation $\lim_{\delta \rightarrow 0^+} \rho(\delta) = 1 - U$ is recovered as the limiting case. This estimate anchors the subsequent optimization but is allowed to move during the likelihood fit.

Step 3 — Initialize $\{b_i, W_{ij}\}$. Given $\tilde{\tau}_{\text{rec}}^{(0)}$ and $\tilde{U}^{(0)}$, compute the latent trajectories $\{x_{j,t}\}$, $\{u_{j,t}\}$, and $\{h_{j,t}\}$ by Eqs. (5), (6), and (13). Initialize the baseline log-odds as:

$$b_i^{(0)} = \log \frac{\bar{p}_i}{1 - \bar{p}_i}, \quad (25)$$

where $\bar{p}_i = T^{-1} \sum_t s_{i,t}$ is the observed spike probability per bin. Initialize all $W_{ij} = 0$, consistent with sparse connectivity and ensuring that early iterations are driven by data rather than random recurrent input.

Step 4 — Burn-in protection. Regardless of the chosen $\tilde{\tau}_{\text{rec}}^{(0)}$, set $x_{j,0} = 1$ for all j (fully recovered) and exclude the first $T_{\text{burn}} = 5\tilde{\tau}_{\text{rec}}^{(0)}/\Delta t$ bins from all likelihood evaluations while retaining them for the forward state computation. This removes dependence on the initial condition, which decays with timescale τ_{rec} .

5.2 Main Optimization: Three-Block Coordinate Descent

With $\{b_i^{(0)}, W_{ij}^{(0)}, \tilde{U}^{(0)}, \tilde{\tau}_{\text{rec}}^{(0)}\}$ initialized, the algorithm alternates between three blocks until convergence. The latent state $x_j(t)$ is not an independent free parameter: given the observed spike train and a candidate pair $(\tilde{U}, \tilde{\tau}_{\text{rec}})$, it is determined by the forward recursion in Eq. (5). Therefore x , u , and h are recomputed whenever either $\tilde{U}$ or $\tilde{\tau}_{\text{rec}}$ changes.

Block A — Update $\{b_i, W_{ij}\}$ with $\tilde{U}$ and $\tilde{\tau}_{\text{rec}}$ fixed. Given the current $(\tilde{U}, \tilde{\tau}_{\text{rec}})$, compute $\{x_{j,t}\}$, $\{u_{j,t}\}$, and $\{h_{j,t}\}$ in a single forward pass. The log-likelihood then decomposes over postsynaptic neurons:

$$\ell(\{b_i, W_{ij}\}) = \sum_{i=1}^N \sum_{t=T_{\text{burn}}}^{T-1} [s_{i,t} \log p_{i,t} + (1 - s_{i,t}) \log(1 - p_{i,t})], \quad (26)$$

where $p_{i,t} = \sigma(b_i + \sum_{j \neq i} W_{ij} h_{j,t})$. This separates into N independent logistic regression problems, one per postsynaptic neuron i , each with at most $N - 1$ predictors $\{h_{j,t}\}_{j \neq i}$. Each is solved with an L1-regularized logistic-regression solver to promote sparsity, exploiting the fact that the true connectivity is approximately 15% dense with unknown support. The N subproblems are fully independent and trivially parallelized across neurons.

Block B — Update $\tilde{U}$ with $\{b_i, W_{ij}\}$ and $\tilde{\tau}_{\text{rec}}$ fixed. The log-likelihood as a function of $\tilde{U}$ is one-dimensional but nonlinear because $\tilde{U}$ changes both the depletion recursion and the release amplitude. For each candidate $\tilde{U} \in (0, 1]$, recompute $\{x, u, h\}$ and evaluate the Bernoulli likelihood. The update can be performed with a bounded scalar optimizer or by automatic differentiation through the forward pass. A weak prior or bounded search interval around the tight-pair initializer can be used to stabilize the known $\tilde{U}$ - W_{ij} scaling coupling.

Block C — Update $\tilde{\tau}_{\text{rec}}$ with $\{b_i, W_{ij}\}$ and $\tilde{U}$ fixed. The log-likelihood as a function of $\tilde{\tau}_{\text{rec}}$ is also one-dimensional and smooth but nonlinear. For each candidate $\tilde{\tau}_{\text{rec}} > 0$, recompute $\{x, u, h\}$ through Eq. (5); then update $\tilde{\tau}_{\text{rec}}$ with a scalar constrained optimizer or automatic differentiation through the forward pass.

Convergence. Outer iterations cycle through Blocks A–C. To avoid early-stage bias from poor STD parameters corrupting the W_{ij} fit, it is useful to hold $\tilde{U}$ and $\tilde{\tau}_{\text{rec}}$ fixed for the first 2–3 outer iterations while

$\{b_i, W_{ij}\}$ stabilize, then release the scalar STD parameters for joint refinement. Convergence is declared when the relative change in log-likelihood between outer iterations falls below a prescribed tolerance, typically 10^{-4} .

5.3 Computational Scaling

The forward pass computing $\{x_{j,t}, u_{j,t}, h_{j,t}\}$ is $\mathcal{O}(NT)$ and fully vectorized across neurons, requiring a single sequential scan over T bins. At $N = 500$ and $T = 10^6$ this amounts to 5×10^8 operations per outer iteration, feasible in seconds on a GPU using `jax.lax.scan` or an equivalent. The Block A logistic regressions involve N problems each of dimension $(N - 1) \times T$, but the L1 penalty and true sparsity of W_{ij} mean that only $\sim 0.15N$ predictors carry nonzero weight, reducing the effective problem size substantially. Blocks B and C are scalar optimizations; each likelihood or gradient evaluation costs one $\mathcal{O}(NT)$ forward pass and is usually not the bottleneck.

5.4 Identifiability

The strongest coupling is between $\tilde{U}$ and W_{ij} : increasing $\tilde{U}$ increases release amplitude and can be partially offset by smaller weights. The tight-pair initializer, the bounded domain $0 < \tilde{U} \leq 1$, and optional weak regularization on $\tilde{U}$ help anchor this direction. A second coupling exists between $\tilde{\tau}_{\text{rec}}$ and W_{ij} : longer recovery times produce more depleted $x_{j,t}$ trajectories, which logistic regression can partially compensate by inflating W_{ij} . These couplings are not exact degeneracies because the model must also match the time course of recovery after bursts, not only the mean coupling level. The recommended practice of fitting $\{b_i, W_{ij}\}$ first and then releasing $\tilde{U}$ and $\tilde{\tau}_{\text{rec}}$ exploits this structure by ensuring the scalar-parameter gradients are evaluated against a stable weight matrix.

References

- [1] M. V. Tsodyks and H. Markram. The neural code between neocortical pyramidal neurons depends on neurotransmitter release probability. *Proceedings of the National Academy of Sciences*, 94(2):719–723, 1997. doi:10.1073/pnas.94.2.719.